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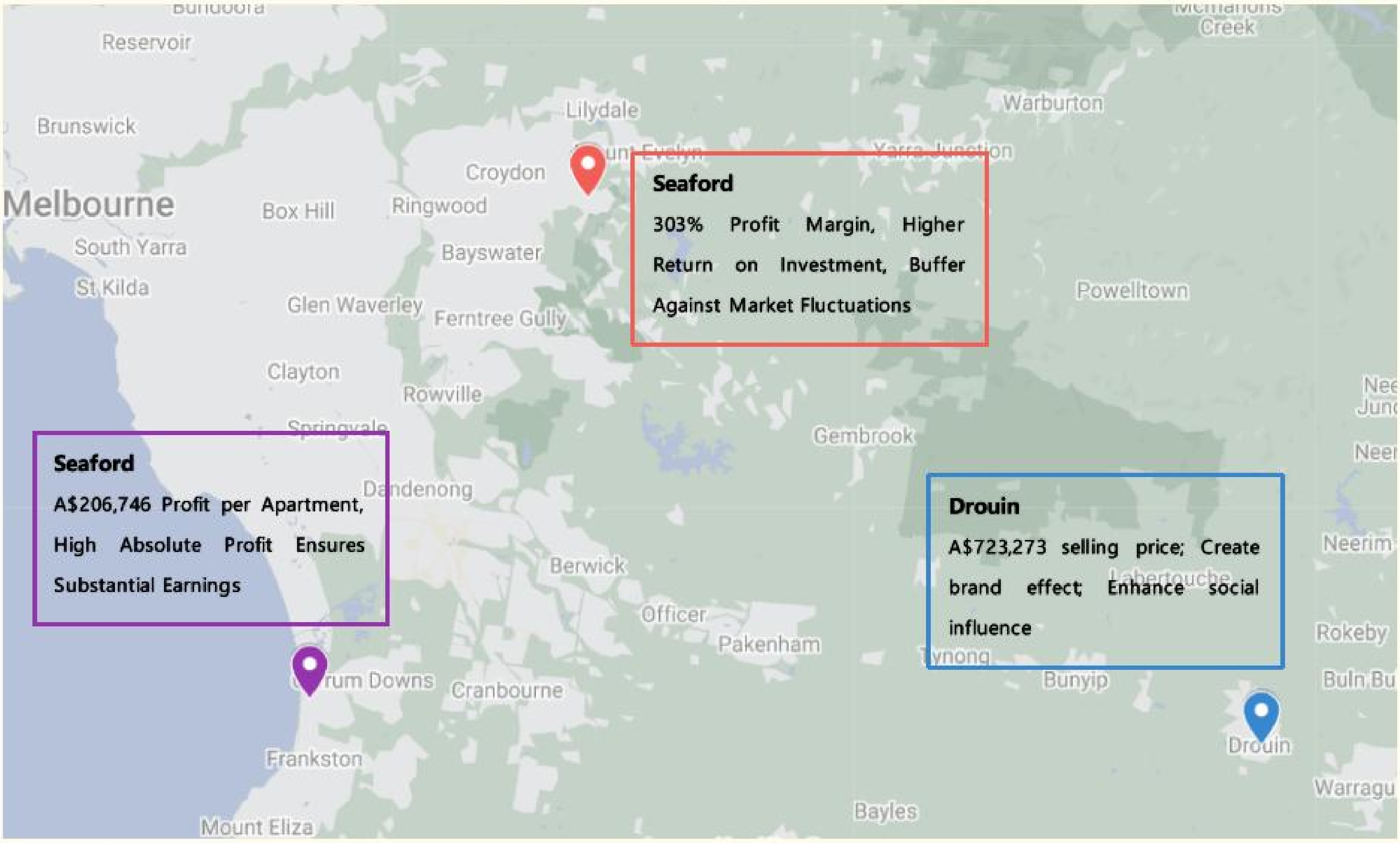


Investment strategy	Investment target
maximum profit margin	**MONTROSE(303%)**
maximum profit per apartment	**SEAFORD(A$ 206,746)**
maximum selling price per apartment	**DROUIN(A$ 723,273)**

See section 3.4 for detailed analysis

Introduction

Background:

The company is looking to invest in and develop new apartments in the state of Victoria, Australia. As a data scientist, I need to **analyse 3 datasets to help company identify where will be a good location to invest**

Objective

- **Provide a recommendation on the best suburb to invest in for next year**.
  - **Estimate a linear regression** with good-fit
  - **Define** what is the **best**
  - **Find a best suburb** according to analysis

Assumption:

- Assuming **past year** in datasets is 2023 and **next year** is 2024
- Company can invest apartments in **only 1 specific suburb around median price in 2023**
- Assuming **unemployment rate = Number of Unemployed People / Labor Force * 100%,** which can not be less than 0

Part 1 - Data Handling & Visualization

1.1 Data Overview

Data structure

- There are 3 datasets with **Primary key(Suburb_name)**, which contains **458 suburbs**. Provide **historical and projected** data on **population growth**, **median income**, **unemployment rates**, and **priority growth areas** respectively. Additionally, **historical median price**, used for understanding past trends and making future projections.
- There are 3 data issues:
  - Class Type Error **in Median_price_2023** Column
  - NA Values in **Historical_population_growth** Column
  - Numeric Error in **Historical_unemployment_rate** Column
- Merge data

	Suburb_name	Median_price_2023	Historical_population_growth	Historical_median_income	Historical_unemployment_rate
1	Length:458	Min. : 64894	Min. :2.050	Min. : 67160	Min. :-3.050
2	Class :character	1st Qu.: 234216	1st Qu.:3.812	1st Qu.:102208	1st Qu.: 3.197
3	Mode :character	Median : 308810	Median :4.415	Median :134793	Median : 5.320
4	NA	Mean : 329789	Mean :4.576	Mean :134272	Mean : 5.551
5	NA	3rd Qu.: 393465	3rd Qu.:5.197	3rd Qu.:164756	3rd Qu.: 7.728
6	NA	Max. :4202660	Max. :8.690	Max. :203191	Max. :17.330

storical_priority_growth_area	Projected_population_growth	Projected_median_income	Projected_unemployment_rate	Projected_priority_growth_
Min. :0.0000	Min. :2.116	Min. : 69167	Min. :-3.133	Min. :0.0000
1st Qu.:0.0000	1st Qu.:3.925	1st Qu.:106170	1st Qu.: 3.172	1st Qu.:0.0000
Median :0.0000	Median :4.534	Median :139586	Median : 5.285	Median :0.0000
Mean :0.1878	Mean :4.717	Mean :139478	Mean : 5.553	Mean :0.1878
3rd Qu.:0.0000	3rd Qu.:5.326	3rd Qu.:171284	3rd Qu.: 7.787	3rd Qu.:0.0000
Max. :1.0000	Max. :8.857	Max. :209992	Max. :17.080	Max. :1.0000

1.2 Data cleaning

Handling Missing Values

- There is a NA in historical demographic data(**historical_median_income**: POINT LONSDALE). One possible way is to fill missing 'Historical median income' with mean($134,322). But I think this data could be accessed online and will be more reasonable. From Australian Bureau of Statistics[i], we find:
  - 2021 Census All persons QuickStats: Median weekly incomes(Family) - $1,994
  - VIC State wage growth: 3.4% in 2022; 3.7% in 2023

  Hence Estimated historical_median_income(POINT LONSDALE, 2023) = $1,994 * (1+3.4%) * (1+3.7%) * 52weeks = $111,180

Correcting Data Types & Values

- Find **309334x** in **Median_price_2023**, Since the datasets is not very large and the data appears to be credible after removing the extraneous character, we will retain the value as **309334**
- Hence change data type of **Median_price_2023** to Quantitative(continuous)
- Since **unemployment rate** can not be less than 0, we may take one of following possible steps based on result of next part (distribution plot):
  - Remove all negative **Historical_unemployment_rate** instances
  - Perform **clipping** on the negative values in the **Historical_unemployment_rate**

Handling Outliers

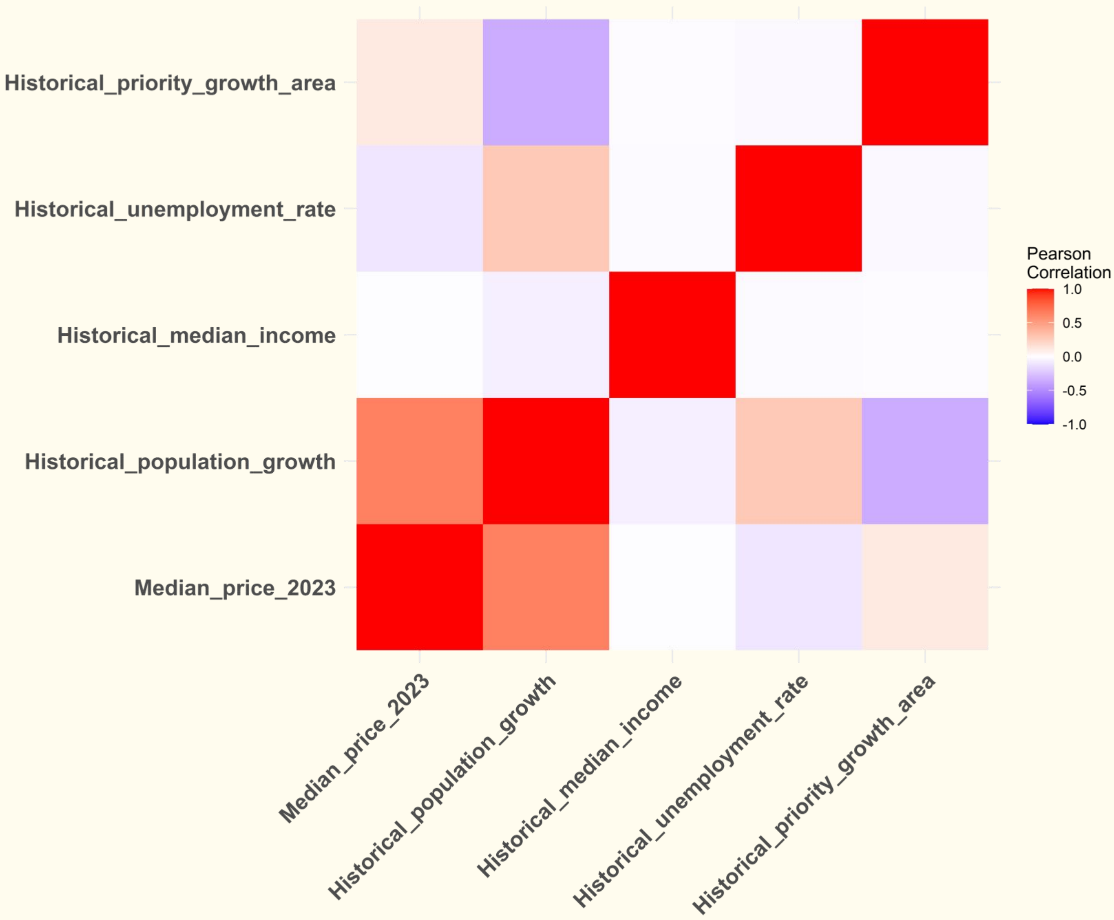
- For Median_price_2023, There is an extreme outlier(value=4202660)
- remove this instance(Suburb_name=MAFFRA)
- Other points seems good(according to 3 times the interquartile range method)

1.3 Data preprocessing & Dive into data

Relationship

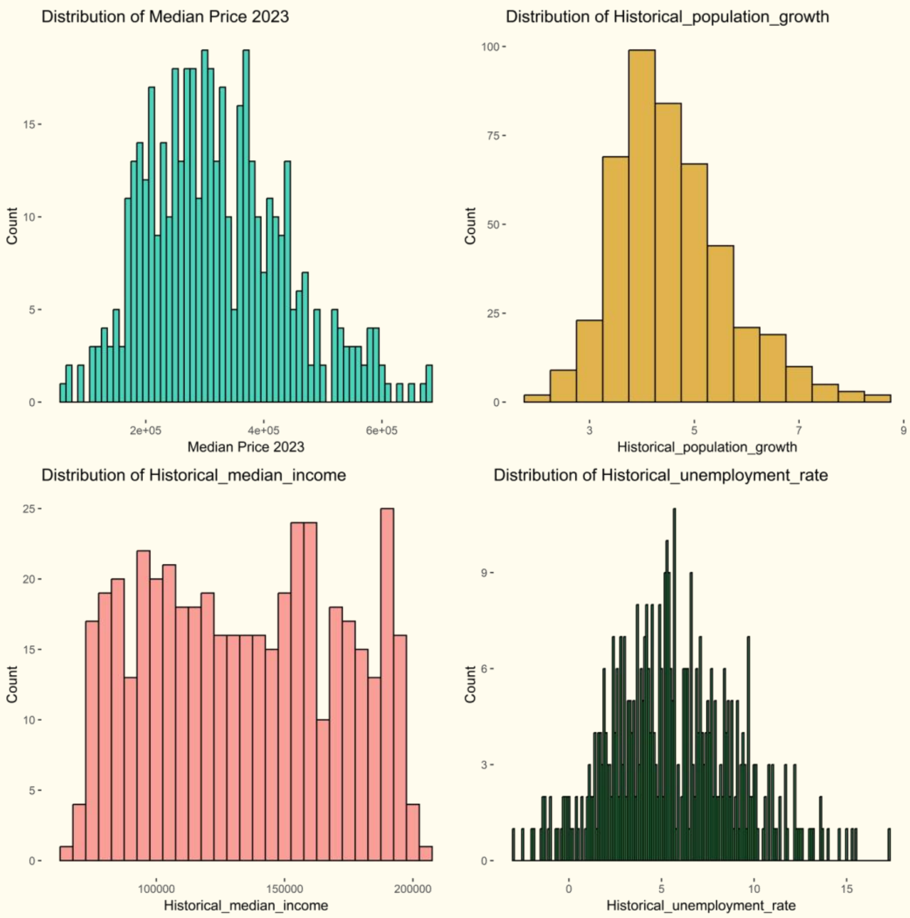
- The scatter plots show strong linear correlation(Positive) between **Historical_population_growth** and **Median_price_2023**
- The scatter plots also show there is no correlation between **Historical_median_income** and **Median_price_2023**

Correction Matrix



- The correlation matrix show strong correlations between **Historical_population_growth** and **Median_price_2023,** there are also some correlations between **Historical_unemployment_rate/ Historical_priority_growth_area** and **Historical_population_growth**. Which is consistent with scatter plots
- No issue of **multicollinearity** was found(correlation values of independent variables are small)

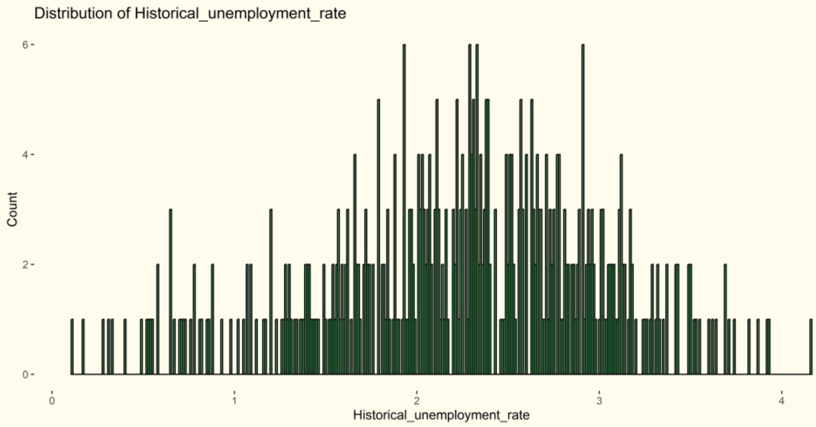
Distribution plot



- Distribution of **Median_price_2023, Historical_population_growth** and **Historical_unemployment_rate** resemble normal distributions
- Perform **clipping** on the negative values in the **Historical_unemployment_rate**, then **adding Small Random Positive Numbers** (Noise: numbers between 0 and 0.8) to Zero Values. The distribution plot of **Historical_unemployment_rate** become:
  - Some accuracy of **Historical_unemployment_rate** has been discarded, but other more effective information such as **Historical_population_growth** has been retained

Data Transformation

- Use square root transformation to reduce the skewness of the data, making it more suitable for linear regression, after transforming:



Data Splitting

- Split the data into training and testing sets. Now we get 3 parts of data.

Train data	365 obs. Of 5 variables
Test data	92 obs. Of 5 variables
New_data	457 obs. Of 4 variables

Part 2 - Model Estimation

2.1 Model Training

Based on analysis before, we try 3 models:

base_model	$P = \beta_0 + \beta_1 X + \beta_2 Y + \beta_3 Z + \beta_4 V + \varepsilon_t$	Base model
removing_model	$P = \beta_0 + \beta_1 Y + \beta_2 Z + \beta_3 V + \varepsilon_t$	Removing **Historical_median_income** according to correlation values
trans_model	$P = \beta_0 + \beta_1 Y + \beta_2 \sqrt{Z} + \beta_3 V + \varepsilon_t$	Based on analysis on data transformation part

- β_i represent the intercepts and the regression coefficients associated with the respective explanatory variables
- ε_t is a disturbance term for quarter t, where $\varepsilon_t \sim WN(0, \sigma^2)$
- X - **Historical median income**; Y - **Historical population growth**; Z - **Historical unemployment rate**; V - **Historical priority growth area**; P - **Predicted Median Price**

2.2 Model Evaluation

Summary

Regression Models Summary

	Dependent variable:		
	Median Price 2023		
	Base Model	Removing Model	Transformed Model
	(1)	(2)	(3)
Historical Population Growth	95,595.67***	95,256.39***	93,926.23***
	(3,756.41)	(3,746.14)	(3,806.60)
Historical Median Income	0.11		
	(0.10)		
Historical Unemployment Rate	-11,840.31***	-11,852.76***	
	(1,139.60)	(1,140.02)	
Transformed Historical Unemployment Rate			-47,907.34***
			(5,062.22)
Historical Priority Growth Area	123,467.90***	122,923.70***	119,765.40***
	(10,266.88)	(10,260.01)	(10,446.15)
Constant	-84,513.01***	-68,085.72***	-20,682.78
	(22,735.19)	(17,587.56)	(19,077.80)
Observations	365	365	365
R^2	0.65	0.65	0.63
Adjusted R^2	0.64	0.64	0.63
Note:			*p<0.1; **p<0.05; ***p<0.01

- Based on summary of 3 models, we find:
  - Coefficient of **Historical_median_income** is not significant at the 0.05 level (p-value > 0.05) in **base_model**, indicating it might not have a strong effect in the model
  - Constant of **trans_model** is not significant at the 0.05 level (p-value > 0.05)
  - All other variables are significant at the 0.001 level

Model Comparison

	Model	MSE_Train	MSE_Test	R2_Train	R2_Test
1	base_model	4721357549	5874409650	0.6486751	0.6337815
2	removing_model	4738387137	5923387303	0.6474079	0.6301234
3	trans_model	4933313302	5974359685	0.6329031	0.6248527

- $MSE = \frac{1}{n}\sum_{i=1}^{n}(P_i - \widehat{P}_i)$; $R^2 = \frac{\sum_i(\widehat{P}_i - \bar{P}_i)^2}{\sum_i(P_i - \bar{P}_i)^2}$
- **Removing_model** has lower Test MSE and higher Test R^2

2.3 Model Selection

Choose **Removing_model** as final model based on reasons below**:**

- All coefficients are statistical significance with lower Test MSE and higher Test R^2
- Provide better balance between model complexity and predictive accuracy on new data
- From Economic perspective, the median family income have not shown statistical significant in the similar models[ii] , which is consistence with our analysis

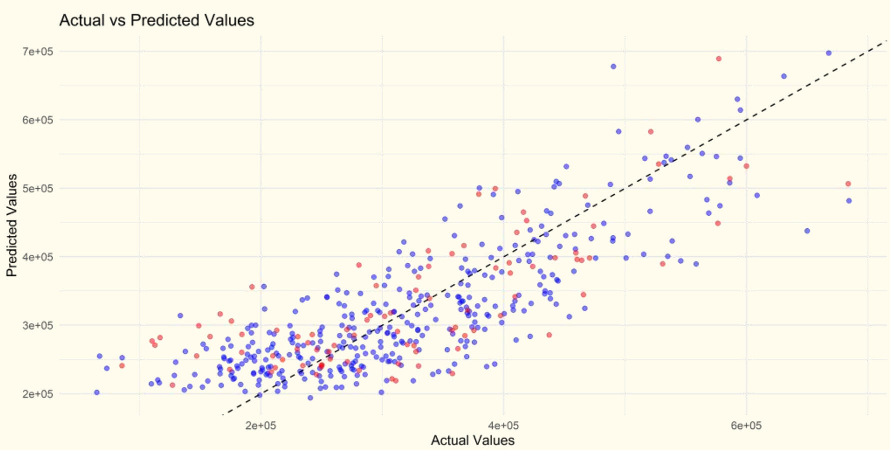
Hence final model is:

$$\mathbf{P} = -68086 + 95256\mathbf{Y} - 11853\mathbf{Z} + 122924\mathbf{V}$$

- Y - **Historical population_growth**; Z - **Historical unemployment_ rate**;

V - **Historical priority growth area**; P - **Predicted Median Price**

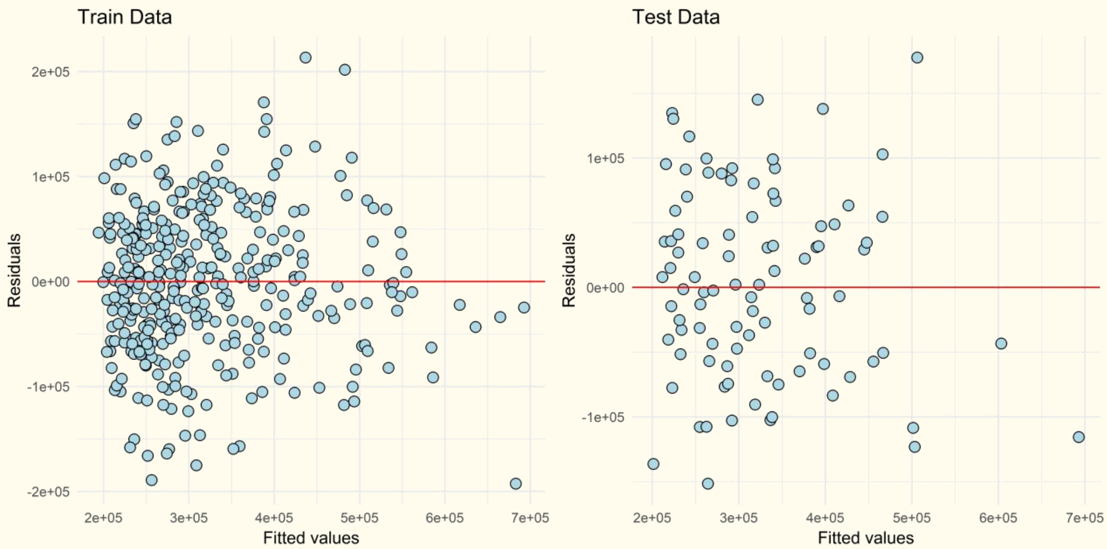
Actual vs Predicted Scatter Plot



Part 3 - Model Interpretation

3.1 Residual analysis

Residuals vs Fitted Plot



- Residuals randomly scattered around the horizontal axis (y = 0) without any discernible pattern and be centered around zero. Also there is no heteroscedasticity

Histogram of Residuals

- Histogram of residuals resemble normal distributions. Combine analysis above, linear model basically satisfies the Gauss-Markov assumptions

3.2 Interpretation & Implications

- The model explained approximately 64.74% of the variance in median prices ($R^2 = 0.6474$), indicating a relatively good fit.
- All predictors were statistically significant ($p < 0.001$), showing strong relationships between these historical factors and median prices.
- Implications of the Coefficients:
  - **Historical Population Growth**: Holding all other independent variables constant, for every one-unit increase in **Historical population growth**, the **Predicted Median price** is expected to increase by 95256.
  - **Historical Unemployment Rate**: Holding all other independent variables constant, for every one-unit increase in **Historical Unemployment Rate**, the **Predicted Median price** is expected to decrease by 11853.
  - **Historical Priority Growth Area**: Holding all other independent variables constant, for every one-unit increase in **Historical Priority Growth Area**, the **Predicted Median price** is expected to increase by 122924.

3.3 Prediction

- Perform **clipping** on the negative values in the **Projected_unemployment_rate,** Then use **Removing_model** make predictions

3.4 Limitations(Based on Gauss-Markov)

1. **Linear in parameters**: The correlation between **Historical_unemployment_rate/ Historical_priority_growth_area** and **Historical_population_growth** may not linear, If there is a nonlinear relationship in the model, the linearity assumption is violated
2. **Random Sampling**: If the sample is not randomly selected, the results may be biased

3. **No Perfect collinearity**: Although correlation values of independent variables are small, The correlation between **Historical_unemployment_rate** and **Historical_priority_growth_area** may reflecting the potential of this issue
4. **Zero conditional mean**: If the error term is correlated with the independent variables, this assumption is violated, leading to biased estimates
5. **Homoskedasticity**: If heteroskedasticity is present, standard errors are unreliable, affecting hypothesis tests and confidence intervals

3.5 Investment suggestion based on model

Define **best suburb to invest** in 3 different ways below

1. A suburb which brings maximum profit margin(per A$)
   - $Profit\ Margin\ = \frac{Predicted\_Median\_price\_2024 - Median\_price\_2023}{Median\_price\_2023}$
   - Advantage: Higher Return on Investment; Buffer Against Market Fluctuations
   - Choose **MONTROSE** as the investment target, with estimated 303% profit margin, alternative suburb - **WARRNAMBOOL**(242%); **COBRAM**(242%)
2. A suburb which brings maximum profit per apartment
   - $Profit\ per\ apartment = \ Predicted\_Median\_price\_2024 - Median\_price\_2023$
   - Advantage: High absolute profit ensures substantial earnings from each property sale
   - Choose **SEAFORD** as the investment target, with estimated A$206,746 profit per apartment, alternative suburb - **MONTROSE**(A$203,786); **WERRIBEE SOUTH**(A$190,575)
3. A suburb which brings maximum selling price per apartment
   - Buy relatively most expensive apartment
   - Advantage: Create brand effect; Enhance social influence;
   - Choose **DROUIN**(A$723,273) as the investment target, alternative suburb -

   SANDRINGHAM(A$709,145); **SEAFORD**(A$697,157)

Investment Plot